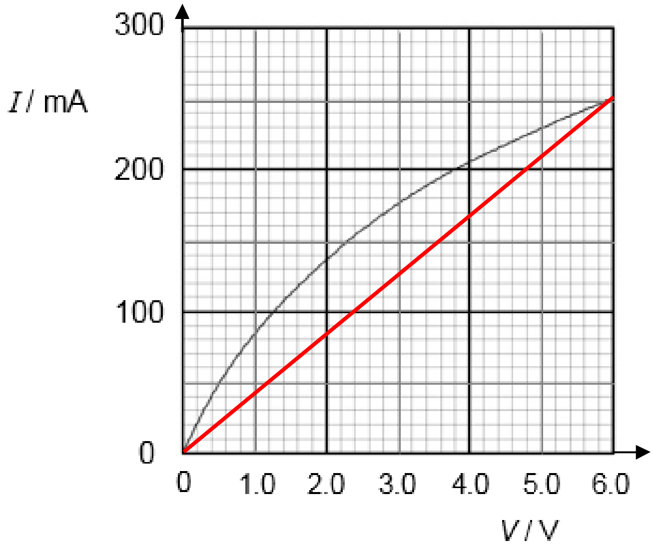
	frequency / Hz	
5ai	As V increases, power dissipation increases, which increases the temperature of the filament. The amplitude of vibration of lattice ions/atomic core increases, causing electrons to collide more frequently with the lattice ions/atomic core as they drift along the filament. This reduces the mean drift speed of the electrons. Hence resistance increases.	B1 B1 B1
5aii	 Straight line passing through origin and (6.0, 250)	B1
5aiii	$\text{change in resistance} = \dfrac{6.0}{0.250} - \dfrac{4.0}{0.205}$ $= 4.5\ \Omega$	C1 A1
5bi	The effective resistance of the LDR and lamp is less than 10 Ω in daylight. Using the potential divider principle, potential difference across the lamp (and LDR) will be less than 6 V (or less than 4 V).	B1 B1

5bii	$R_{\text{lamp}} = \dfrac{V^2}{P_{\text{lamp}}} = \dfrac{6.0^2}{1.5} = 24\ \Omega$	C1
	For V across lamp to be 6 V, $R_{\text{effective}} = 20\ \Omega$	
	$\dfrac{1}{R_{\text{effective}}} = \dfrac{1}{R_{\text{LDR}}} + \dfrac{1}{24}$	
	$\dfrac{1}{20} = \dfrac{1}{R_{\text{LDR}}} + \dfrac{1}{24}$	C1
	$R_{\text{LDR}} = 120\ \Omega$	A1